

SPECIAL PROJECT SPECIFICATION

SECTION 401 ASPHALT BASE, WEARING, AND PATCHING AND LEVELING COURSES

401.1-DESCRIPTION:

This work shall consist of constructing one or more courses of asphalt concrete, mixed mechanically in a plant, composed of aggregate and asphalt material designed in accordance with the Marshall design system as specified in the contract documents, on a prepared foundation in accordance with these specifications and in reasonable close conformity with the lines, grades, weights or thicknesses, and cross sections shown on the Plans or established by the COR. The unit of measurement for asphalt concrete will be by the ton, square yard or cubic yard. The work will be accepted in accordance with these Specifications and the applicable requirements of Sections 104, 106, and 109.

401.2-MATERIALS:

The materials shall conform to the following requirements:

Coarse Aggregate	703.1 thru 703.3 (Note 1 & Note 2)
Fine Aggregate	702.3
Mineral Filler	702.4
Performance Graded Binders	705.5

Note 1 The total Shale, coal, and other lightweight deleterious material and friable particles shall not exceed 3%

Note 2 When slag is specified in the contract, the coarse aggregate shall be slag which meets the requirements of 703.3, except as amended in this subsection.

401.2.1-Performance Graded Binder Grade: The standard grade for Performance Graded (PG) binders shall be PG 64S-22. All mix designs and incorporated materials shall be from a WVDOH approved source.

CONSTRUCTION METHODS

401.3-GENERAL:

Construction methods to be used in performing the work shall be submitted to the COR for review prior to the start of work. This review may require modification of the proposed methods to provide the desired end product. All equipment, tools, machinery, and plants shall be maintained in a satisfactory working condition.

401.4-COMPOSITION OF MIXTURES:

401.4.1-General:

The aggregate for use in the designated mixture shall consist of a mixture of aggregate (coarse, fine, or mixture thereof) and mineral filler if required. RAP shall not be used in the mix.

401.4.2-Job Mix Formula: Job Mix Formula (JMF) is the specification for a single mix produced at a single plant. This mix may be specific to a single project or be used on multiple projects if the basic design criteria (design compaction level and PG Binder grade) are the same.

The Contractor shall submit a proposed JMF for each combination of aggregate and asphalt material for each type of asphalt concrete to be produced. Depending on the design type, the JMF gradations shall be within the tolerances set forth in either Table 401.4.2A. Marshall mix designs shall be developed in accordance with WVDOH MP 401.02.22.

Each proposed JMF must be documented and the entire JMF package shall be forwarded for review to the Contracting Officer's Representative (COR). If the JMF requires revision, it will be returned to the designer.

The Documentation shall contain the following information:

- i. Identification of the source and type of materials used in the design.
- ii. The aggregate blend percentages and the percentage for each sieve fraction of aggregate considered the desirable target for that fraction.
- iii. The percentage of asphalt binder representing the optimum asphalt content for the JMF submitted, which is to be considered the desirable target percentage.
- iv. The temperature of the completed mixture at the plant which shall be within $\pm 25^{\circ}\text{F}$ of the median mix temperature established by the temperature-viscosity chart or as recommended by the asphalt supplier.
- v. The ratio (calculated to the nearest one-tenth percent) of the Fines to Asphalt (FA). For Marshall mixes the ratio is defined as the percentage of aggregate passing the No. 200 (75 μm) sieve, divided by the percentage of asphalt content calculated at the percentage optimum asphalt content of the design.

TABLE 401.4.2A
Design Aggregate Gradation Requirements for Marshall Mix Designs

Type of Mix	BASE-I	BASE-II (Patch & Level)	WEARING-I (Scratch-I)	WEARING-III (Scratch-III)
	Nominal Maximum Size			
Sieve Size	1 ½" (37.5 mm)	¾" (19 mm)	3/8" (9.5 mm)	No. 4 (4.75 mm)
2" (50 mm)	100	-	-	-
1 ½" (37.5 mm)	90-100	-	-	-
1" (25 mm)	90 max	100	-	-
¾" (19 mm)	-	90-100	-	-
1/2" (12.5 mm)	-	90 max	100	-
3/8" (9.5 mm)	-	-	85-100	100
No. 4 (4.75 mm)	-	-	80 max	90-100
No. 8 (2.36 mm)	15-36	20-50	30-55	90 max
No. 16 (1.18mm)	-	-	-	40-65
No. 30 (600 um)	-	-	-	-
No. 50 (300 um)	-	-	-	-
No. 200 (75 um)	1-6	2-8	2-9	3-11

If it becomes necessary to change aggregate sources, a new mix design shall be developed and submitted for approval. When using neat (unmodified) PG Binders, the binder source may be changed without requiring a new mix design, but the binder grade must always remain the same for each design. If a source change results in the use of an additive- enhanced modified binder of the same grade, a new mix design will be required.

If a modified binder source is changed or if the modification process is changed, a new mix design shall be developed and submitted for approval. A source change to a new location with the original manufacturer/supplier and the original modification process will not require a new mix design.

At no time shall different grades of PG Binders be mixed together in the same storage tank. When it is necessary to switch to a new binder grade the tank shall be drawn down as far as possible, normally to the top of heating coils, before refilling with the new binder. The new binder shall be circulated thoroughly before restarting production.

401.4.2.1 - Warm Mix Asphalt: Warm Mix Asphalt shall not be used on this project.

401.5-TESTING:

401.5.1-Test Methods:

AASHTO T168	Sampling Hot-Mix Asphalt
AASHTO T11	Materials Finer than NO. 200 Sieve in Mineral Aggregates by Washing
AASHTO T27	Sieve Analysis of Fine and Coarse Aggregates
AASHTO T30	Mechanical Analysis of Extracted Aggregate
AASHTO T164	Quantitative Extraction of Bitumen from Bituminous Paving Mixtures
AASHTO T308	Asphalt Content of HMA by the Ignition Method (Test Method A)
AASHTO T245	Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus
ASTM D5581	Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus (For Base-1 Marshall designs only)
AASHTO T166	Bulk Specific Gravity of Compacted Bituminous Mixtures
AASHTO T209	Maximum Specific Gravity of Bituminous Paving Mixtures

401.6-CONTRACTORS QUALITY CONTROL:

401.6.1-Quality Control Testing:

Quality control of bituminous asphalt is the responsibility of the Contractor. The Contractor shall maintain equipment and qualified personnel including at least one certified Asphalt Plant Technician at each plant. The technician shall be in charge of all plant quality control activities such as mix proportioning and adjustment and all sampling and testing activities necessary to maintain the various properties of asphalt within the limits of the specification.

The Contractor shall maintain necessary equipment and qualified personnel including at least one certified Asphalt Field and Compaction Technician at each project during paving operations. Additionally, a certified Asphalt Field and Compaction Technician with certification to perform nuclear density testing of asphalt pavements shall perform

all testing necessary to assure compaction of the asphalt meets specification requirements.

The Contractor, shall design a workable Quality Control Plan, detailing the type and frequency of sampling and testing deemed necessary to measure and control the magnitude of the various properties of the asphalt governed by these specifications. This plan, prepared in accordance with WVDOH MP 401.03.50 shall be submitted to the COR for review prior to production of material under this specification.

401.6.2-Job Mix Formula (JMF) Field Verification: A proven and WVDOH field verified Marshall Designed JMF shall be used for all paving applications. A copy of the JMF and proof of field verification shall be provided to the COR prior to placement of any HMA material. If a new JMF is utilized on the project site it shall be verified during the first days of plant production. The verification shall be in accordance with the guidelines established in WVDOH MP 401.02.27. The field verification is for the purpose of demonstrating that the JMF can be produced within the specified tolerances set forth in the MP. If the mix cannot be produced within these requirements, a new mix design will be required.

401.6.3-Quality Control Testing Requirements: Test requirements for quality control shall be as set forth in WVDOH MP 402.02.27 for Marshall designs. If it is determined that a mix cannot be consistently produced within the tolerance limits of the specified design properties, approval of the mix may be revoked, and the contractor will be required to provide a new mix design.

401.6.4-Compaction: Projects shall be tested for compaction in accordance with the Rollerpass test method described in Section 401.6.4.2.

In addition, areas of Trench Paving, Pavement Widening, and Pavement Repairs shall be tested in accordance with the Rollerpass test method or to the satisfaction of the Engineer.

When asphalt is placed in areas that require a non-uniform thickness and tapers to a thin edge including patch and level and scratch, acceptance testing of the material is not required. Compaction shall be accomplished with a minimum of eight (8) roller passes prior to reaching the temperature specified in section 401.10.4. A pass shall be defined as the entire roller traversing a spot on the pavement. Compaction shall be performed by a three-wheel (steel) roller or pneumatic-tire roller.

Acceptance testing is not required on areas which are too narrow to be compacted with the roller use for the mainline paving, as defined in the contractor's QC plan; such areas shall be compacted to the satisfaction of the Engineer.

401.6.4.1-Density Testing: All Gauge standardization procedures, calibration procedures and all Density testing conducted shall be in accordance with AASHTO T355 - *Standard Method of Test for In-Place Density of Asphalt Mixtures by Nuclear Methods*.

Testing shall be conducted in the backscatter position and follow AASHTO T355 with the exception that no gauge rotation will be required. All gauge tests shall be conducted with the source end of the gauge in the direction of paving.

401.6.4.2-Roller Pass Testing: A Rollerpass Control Section shall be completed on a daily basis, when roadway conditions change where they would affect the compaction effort, or when the Engineer determines the current rollerpass is unsatisfactory. A rollerpass shall be established prior to the mat reaching the temperature specified in section 401.10.4.

A rollerpass shall be conducted in the following manor:

1. The Rollerpass Control Section shall be conducted 100 feet beyond the initial transverse construction joint. If an additional rollerpass is required by the Engineer conduct it immediately.
2. Apply four passes with the breakdown roller (a pass shall be defined as the entire roller traversing a spot on the pavement) to the roadway, then conduct two randomly located nuclear density testing within the section; record the results, the average, and the mat temperature at each test location.
3. Apply an additional two passes and repeat the nuclear density testing in the same locations; record the results, the average, and the mat temperature at each test location.
4. Repeat step 3 until one or more of the following conditions occur: (a) less than 5 kg/m³ increase occurs between the average of two sets of readings, (b) the density of the material exceeds 97.0% Gmm, (c) one of the two test location “breaks over” (i.e. shows a decrease in density) after exceeding 92.0% of Gmm, or (d) the temperature of the mat has fallen below 175 °F.
5. Compute the Percent of Gmm using the average of the two readings and record on the worksheet.

If the mat begins to show signs of distress (such as excessive surface aggregate breakage or mat cracking) before reaching 175 °F, then discontinue rolling and record the number of roller passes completed before the stress signs occurred. If a tender mix, as defined in Section 3.3, is encountered the Contractor may be allowed to continue rolling at lower temperatures if it can be demonstrated that additional densification can be achieved at a lower temperature without causing any pavement distress. Once the control section is completed, the density shall be equal to or greater than 92.0% of Gmm. If the density does not meet 92% of Gmm, repeat the procedure above immediately. If after two control sections the density still does not meet 92% of Gmm, contractor shall apply the number of passes associated with the highest percent density, minimum of 8 passes, unless the Engineer determines more appropriate means. To help with this decision, an evaluation

may be made of the existing pavement condition and any density test results obtained prior to construction of the test section will be reviewed.

If the density within the control section meets 92% of Gmm conduct a proving section in the proceeding 1000 feet. Within the proving section, apply the established number of passes and conduct five randomly located nuclear density tests. The average of these five tests shall exceed a minimum density of 92% Gmm and be within ± 50.0 kg/m³ of the average wet density determined in the Rollerpass control section. If this is not achieved a new rollerpass section shall be conducted.

All data shall be submitted the Engineer on associated Rollerpass forms.

401.7-ACCEPTANCE TESTING:

401.7.1-Acceptance Testing of HMA: Acceptance of all asphalt material will be based on observation and results of Quality Control testing and results performed and submitted by the contractor. Independent testing may be conducted by the Forest Service for further verification and final acceptance.

401.7.2-Surface Tolerance: It is the intent of these specifications that projects shall be constructed to provide a smooth riding surface. The contractor will take all reasonable actions to ensure that the final driving surface is constructed using industry BMPs and acceptable techniques to provide a smooth final product.

When compaction is completed on the course, it shall present a uniform surface, true line and grade. When tested with a straightedge of approximately 10 feet (3 meters) in length and a template of the specified dimensions, the finished base course shall not show a deviation greater than ¼ inch (6 mm) and the finished wearing course shall not show a deviation from the required surface greater than 3/16 inch (5 mm).

The Contractor shall provide the straightedge and template for checking the surfaces and an employee to use them under the direction of the COR. Any irregularity of the surface exceeding the limits specified shall be corrected. Depressions which develop after the initial rolling shall be corrected by loosening the mixture and adding new material. High places shall be corrected by removing excess material.

Areas of completed courses found to be defective shall be removed and replaced with approved mixtures laid in accordance with these specifications, and no additional compensation will be allowed for materials used or work involved in replacing defective areas.

401.7.3-Compaction: Acceptance testing for compaction shall be performed in accordance with the rollerpass procedure described in WVDOH MP 401.05.20. Acceptance testing is not required on areas in which a full-size roller is restricted from

properly compacting the mat. These areas shall be compacted to the satisfaction of the COR.

401.8-WEATHER RESTRICTIONS:

Bituminous asphalt shall only be placed when the surface is dry and when weather conditions are such that proper handling, finishing, and compaction can be accomplished. In no case however, shall asphalt be placed when the surface temperature is below 50 °F (10 °C).

In addition to the surface temperature requirements, no wearing courses shall be placed when the air temperature is less than 40° F (4° C).

401.8.1-Cold Weather Paving: When the air temperature falls below 50° F (10° C) or the surface temperature falls below 60° F (15° C) the provisions for cold weather paving shall apply.

Paving shall be performed so that the transverse joints in adjacent lanes shall be no more than 50 feet (15 meters) apart at the end of each day's paving operations. The surface temperature shall be taken a minimum of once every hour and shall be taken in all shaded areas. The temperature requirements as listed in section 402.8 shall apply.

It shall be the Contractor's responsibility to monitor the declining surface temperature to insure that material delivery from the plant be terminated so as to allow that all material can be placed prior to the surface temperature falling below the surface temperatures listed in section 401.8.

The temperature of the delivered material and the mat temperature at the time final density is obtained shall be taken for every truck load. These temperatures shall be in accordance with Section 401.10.3 for delivered material and 401.10.4 for the mat temperature at the time final density is obtained.

401.9-EQUIPMENT:

401.9.1-Plants: All plants in West Virginia producing asphalt for the Forest Service shall provide documented evidence of compliance with current requirements of the West Virginia Air Pollution Control Commission.

All plants which are not in West Virginia but producing asphalt for the Forest Service shall provide documented evidence of compliance with current requirements of the laws and regulations of the State in which they are producing, applicable to air pollution. All plants shall meet the general requirements set forth in AASHTO M156 unless it can be demonstrated to the satisfaction of the COR that a consistent quality mix can be produced with modifications to any of these requirements.

401.9.2-Dust Collector: An efficient dust collecting system shall be provided to prevent the loss of fine material. The material collected may be returned to the mixture at a uniform rate or discarded.

401.9.3-Truck Scales: Truck scales shall be provided at each Plant, except that truck scales are not required at properly calibrated automatic batching plant facilities which are equipped with digital printout equipment, and which load the trucks directly from the mixer or the weigh hopper in a surge or storage bin.

A person designated as a weigher shall be provided by the producer. The weigher shall certify that the weight of the asphalt, as determined either by the truck scales or from the digital printout of the batch weights, is correct.

Each truck shall be weighed empty prior to each load, except at automatic batch plants approved to operate without truck scales.

All truck scales shall be mounted on solid foundations which will insure them remaining plumb and level.

Approval and sealing of scales shall be conducted at the frequency determined by the West Virginia Division of Labor, Bureau of Weights and Measures, and when the plant is moved, or upon the request of the COR. The COR shall be notified of any scale malfunctions when material is being furnished to Forest Service projects. The Forest Service may, at its option, accept inspection and sealing by out-of-state agencies when the mixing plant is located outside West Virginia.

A digital recorder shall be required on all truck scales. The digital recorder shall produce a printed record of the gross, tare and net weights, and the time, date, truck identification, and project number. Provisions shall be made for constant zero compensation and further provisions shall be made so that the scales may not be manually manipulated during the printing process. The system shall be interlocked to allow printing only when the scale has come to rest. In case of breakdown of the automatic equipment, the COR may permit manual operation for a reasonable time, normally not to exceed 48 hours, while the equipment is being repaired.

The scales shall be of sufficient size and capacity to weigh the loaded trucks that are used for delivery of asphalt from the plant.

401.9.4-Test Weights: As part of its standard equipment, each plant which proportions aggregate by weight shall provide a minimum of ten 50-pound (22.68 kg) test weights for the purpose of maintaining the continued accuracy of its weighing equipment.

Plants which proportion asphalt material by weight shall furnish, in addition to the above, one five pound (2.268 kg) test weight.

401.9.5-Surge and Storage Bins: During the normal daily operation of the plant, asphalt may be stored in a surge or storage bin for a maximum of 12 hours, provided the bin has received prior evaluation and acceptance through the District plant inspection. The resulting temperature of the material at time of placement and compaction shall be sufficient to comply with 401.10.3 and 401.10.4

Longer silo storage times, up to 24 hours, may be permitted for dense graded asphalt if the storage silo is insulated and/or heated to assure that the proper mix temperature is maintained. The gates at the bottom of the storage silo shall be adequately heated and sealed when the asphalt is held for the extended period of time. An inert gas system may be used to purge the silo of oxygen to prevent oxidation of the asphalt. The asphalt delivered from the storage silo shall meet all of the specification requirements.

When asphalt is stored for the extended time period, it shall not be used until the temperature has been checked and the asphalt has been visually inspected for hardening of the mix and stripping of the asphalt from the aggregate. Approval of the extended storage time may be revoked if it is determined through inspection and/or testing that the extended storage is having a detrimental effect on the asphalt.

Loading of trucks through the storage bin will only be permitted when a minimum 25-ton (23-Mg) buffer of material is being maintained or an amount as recommended by the bin manufacturer, means shall be provided for loading the trucks directly from the mixer when the storage bin is not in operation.

401.9.6-Inspection of Equipment and Plant Operations: The COR shall have access to the plant to assure the adequacy of the equipment in use, to inspect the conditions and operation of the plant, to verify weights, to verify the proportion and character of materials, and to determine if specified temperatures are being maintained in the preparation of the mixture.

401.9.7-Trucks for Transporting Mixture: The inside surfaces of trucks shall be thinly coated with a soapy water or a mixture with not more than ten percent lubricating oil. The use of diesel fuel, kerosene, or similar solvent-based products which can dissolve the asphalt film from the aggregate particles will not be permitted. Any commercial release agent which can be certified as harmless to the mix may be used; however, the Forest Service reserves the right to restrict any release agent that is shown to cause problems during placement of the mix. In the case of mixtures composed of PG Binders which contain polymer modification, truck surfaces should be coated with a release agent recommended by the binder supplier. All excess coating material shall be removed from the truck bed prior to loading the asphalt.

All truck beds shall be insulated with approved material. No trucks shall be used which cause segregation of the materials, which show large oil leaks, or which cause undue delays in delivery of material. All trucks shall be provided with a waterproof cover and a hole in the body for the purpose of conveniently checking the temperature of the load. Covers shall be suspended slightly above the mixture, shall extend over the sides of the

truck, and shall be securely fastened to eliminate air infiltration and to prevent water from coming in contact with the mixture.

401.9.8-Laboratory: A testing facility or laboratory, as described below, shall be provided within reasonable proximity of the asphalt plant. Plant operations must be visible from within the laboratory.

The laboratory shall be of sufficient size to hold all laboratory test equipment and supplies with adequate floor space to allow the technicians to test samples in an efficient manner. The laboratory shall be furnished and maintained with adequate ventilation, heat, light, water, sink and drainage, electrical or gas outlets, or both, work table, shelves, and supply cabinets.

The laboratory shall be supplied with the equipment and materials needed to perform required daily testing and these shall be maintained to meet the applicable requirements of AASHTO or ASTM.

401.9.9-Asphalt Paving Equipment: Asphalt paving equipment shall be self-contained and of sufficient size, power and stability to receive, distribute and strike-off the asphalt mixture at rates and widths commensurate with the typical sections and other details shown on the plans. The paver shall be provided with an activated screed or strike-off assembly equipped to be heated. Approval of the paver by the Engineer will be based on the demonstrated capability of the equipment to place the mixture to the required cross-section, profile and alignment in an acceptable, finished condition ready for compaction.

The paver shall be equipped with means of preventing the segregation of the coarse aggregate particles when moving the mixture from the paver hopper to the paver augers. It shall also be capable of pushing a sufficient amount of the mixture under the auger gearbox to prevent streaking or tearing of the mat. Some paver models may require the installation of a manufacturer retrofit kit or equipment modification to accomplish this.

Specialized equipment or hand methods approved by the Engineer may be employed to spread the asphalt mixture where the use of standard full-scale paver is impractical due to the size or irregularity of the area to be paved.

Pavers shall be equipped with mechanical or automatic grade and slope controls. The use of automatic grade and slope controls with a traveling straight edge shall be required only when specified on the Plans or in the Proposal. Both the grade and slope controls shall be in working order at all times. In the event of failure of the automatic controls, the Contractor will be permitted to finish the day's work using manual controls but will not be allowed to resume work the following day until the grade and slope controls are in proper working order.

401.9.10-Compaction Equipment: Compaction may be performed by self-propelled steel-wheeled, pneumatic-tired and/or vibratory rollers. Hand-held rollers or vibrating plates may be used in small inaccessible areas if approved by the Engineer. Prior to use

on any project, the roller shall be inspected to see that it is in good mechanical condition. The total weight, weight per inch of width (steel-wheeled), and average ground contact pressure (pneumatic-tired) shall be documented.

401.10-PAVING OPERATIONS:

401.10.1-Cleaning and Sweeping: Immediately prior to the arrival of the paving mixture, the existing base or surface shall be thoroughly cleaned by the use of tools and equipment as may be required to remove all mud, dirt, dust, and other caked or loose material foreign to the type of treatment or surface being placed. The cleaning shall be done to a minimum width of one foot on each side beyond the width of the surface being placed.

401.10.2-Patching and Leveling and Scratch Courses:

401.10.2.1-Patching and Leveling: A tack coat shall be applied to the existing pavement prior to placing patching and leveling.

Patching and leveling shall be placed at various locations throughout the project to remove irregularities in the existing pavement such as dips, or to raise the outside edge of the existing pavement to provide a uniform template prior to placing a base or wearing course. Patching and leveling shall not be placed as a continuous layer or course over the full width and length of the project.

Compaction of patching and leveling shall be performed with three-wheel (steel) or pneumatic-tire rollers.

401.10.2.2-Scratch Course: A tack coat shall be applied to the existing pavement prior to placing a scratch course.

The scratch course shall be placed to the limits designated on the plans. Scratch course shall be placed with a paving machine. The paving machine screed shall be set to drag on the high areas of the existing pavement, only depositing material in ruts and other depressions.

The wearing course, or at least one lift of base course, should be placed over the scratch course prior to maintaining traffic in the lane where the scratch course has been placed. All repairs to a scratch course due to traffic damage shall be at the contractor's expense.

Compaction of a scratch course shall be performed with a three-wheel (steel) or pneumatic-tire roller.

401.10.3-Spreading and Finishing: Before spreading any material, the contact surfaces of curbs, gutters, manholes, and of adjacent Portland cement concrete pavement edges shall be painted or sealed with asphalt material. Exact edge of pavement, except on

concrete, shall be established by a string or chalk line for a distance of not less than 500 feet ahead of the spreading operation.

For mixes produced with neat (non-modified) asphalts (which may include PG 70-22, PG 64-22, PG 64-28, and PG 58-28) the temperature of the mixture at the time of placement shall be within the temperature requirements of the JMF. The JMF temperature range shall be within the master temperature range of 250° and 338° F (121° and 170° C) unless otherwise specified by the asphalt supplier. The mix temperature shall be monitored by inserting a dial type thermometer into the mix through the hole in the truck bed.

The temperature of the completed mix, when measured at the plant, shall be within the tolerance as established by the JMF. The first load which demonstrates temperatures outside of that range shall be accepted, provided that the temperature is still within the master temperature range. No additional loads of material shall be run out until necessary steps are taken to reestablish the temperature of the mix within the plant tolerance. When measured at the project site, the temperature of the mix shall be within the tolerance established by the JMF. The first truck load of material which demonstrates temperatures outside of that range or any trucks in transit at that time shall be accepted provided temperatures are within the master temperature range. Any truckload of material which exceeds the master temperature range may be rejected by the COR. However, the plant shall immediately be notified that no additional loads of material are to be dispatched until necessary action is taken to reestablish temperature within JMF specification limits.

When the surface temperature falls to within 10° F (6° C) of the weather restrictions of Section 401.8, the mix temperature may be increased up to a maximum of 338° F (170° C) unless otherwise specified by the asphalt supplier. The temperature of each truckload of material shall be monitored for compliance. Any truckload of material which exceeds this maximum temperature may be rejected by the COR.

Mixes produced with asphalts that contain modifiers for high or low temperature performance enhancement shall meet the temperature requirements recommended by the asphalt supplier, which will be referenced on the JMF.

401.10.4-Rolling Procedure: Shoulders, ramps, and similar areas shall be compacted in the same method as the mainline.

During rolling, roller wheels shall be kept moist with only enough water to avoid picking up material. Fuel oil on roller wheels or pneumatic tires is not allowed. Rollers shall move at a slow but uniform speed with the drive roll or wheels nearest the paver.

If rolling causes material displacement, the affected area shall be loosened at once with lutes or rakes and restored to its original grade with loose material before being re-rolled. Heavy equipment, including rollers, should not be permitted to stand on the finished surface before it has thoroughly cooled or set.

Mat temperature shall be measured using a non-contact infrared thermometer. The required density shall be obtained prior to the mat temperature reaching 175° F (80° C). The Contractor may be allowed to lower this temperature to 165° F (74° C) if they can demonstrate during the first day of placement of each lift on each project that additional densification can be achieved without causing any pavement distress.

401.10.5-Joints: The longitudinal joint in any layer shall offset that in the layer immediately below by approximately six inches; however, the joint in the top layer shall be at the centerline of the pavement if the roadway comprises two lanes of the width, or at lane lines if the roadway is more than two lanes in width. The transverse joint in any layer shall offset that in the layer immediately below by approximately six feet.

All transverse joints between existing and new pavement shall be “heeled-in” to the existing surface at the beginning and at the end of the project and at all other locations where the new pavement terminates against an existing pavement. Transverse joints between one day's production and the next shall be carefully constructed and shall be formed by cutting back into the existing section to expose the full depth of the course. All joints shall be squared up to the full vertical depth of the course to be placed, and a tack coat of asphalt material shall be applied. Joints adjacent to curbs, gutters, or adjoining pavement shall be formed by transporting back sufficient hot material to fill any space left uncovered by the paver.

The longitudinal joint between adjacent mats shall be set up to a sufficient height to receive the full compactive effort from the rollers and shall be tacked prior to placing adjacent material.

Transverse joints shall be checked for smoothness with a ten foot straight edge provided by the Contractor. All surface irregularities shall be corrected prior to proceeding with paving operations.

401.11-PROTECTION OF PAVEMENT AND TRAFFIC CONTROL:

The Contractor shall be responsible for the protection of asphalt surfaces from damage by their equipment and personnel.

401.12-METHOD OF MEASUREMENT: Asphalt concrete will be measured by the ton. The quantity will be determined by the Contractor from the total weigh slips for each vehicle load weighed upon an approved standard scale or from digital printout slips from an automatic batching plant and certified by the Contractor as correct.

Any patching or leveling mixture placed on a subbase or base course constructed in the same Contract with the asphalt items shall be at the expense of the Contractor. No additional compensation will be allowed for the material or any work incidental to its placement.

401.13-BASIS OF PAYMENT:

The quantities determine as provided above, will be paid for at the contract unit price for the items listed below, which prices and payment shall be full compensation for furnishing all the materials and doing all the work herein prescribed in a workmanlike and acceptable manner, including all labor, tools, equipment, field laboratory, supplies, and incidentals necessary to complete the work.

The conditioning, cleaning, and sweeping of the existing base or underlying surface shall be considered as part of the construction of the appropriate items listed in 401.14, and no additional compensation will be allowed for "Cleaning and Sweeping".

There will be no additional compensation for tack coat material used for minor (spot) areas to be patched and leveled; the cost of this tack coat material will be included in the unit bid price for the "patching and leveling" Item.

There will be no additional compensation for Interim Pavement Markings.

401.13.3- When a Lot of asphalt pavement does not meet the density requirements of 401.7.3, the Price shall be adjusted as follows.

FORMULA-1: Projects requiring only mat density testing:

Lot Price Adjustment (Mat only) = (unit price) X (Lot quantity) X (mat density price adjustment % from Table 402.13.3A)

TABLE 402.13.3A

Adjustment of Contract Price for Pavement Density not Within Tolerance of Density	
Percent of Density	Percent of Contract Price to be Paid
Greater than 96%	See Note 1
92% to 96%	100%
91%	98
90%	96
89%	92
88%	88
Less than 88%	88 - 1 per each point of density below 88%

Note 1 Mat density slightly above 96% is normally only a problem if it leads to asphalt flushing on the surface of the mat or rutting due to an unstable mix. The COR will make a special evaluation of the material and determine the appropriate action

401.14-PAY ITEMS: ITEM DESCRIPTION UNIT

401001- “design method” Asphalt Base Course, “aggregate type”, Type “mix type”

401002- “design method” Asphalt Wearing Course, “aggregate type”, Type “mix type”

401003- “design method” Asphalt Patching And Leveling Course, “aggregate type

401007- “design method” Asphalt Scratch Course, “aggregate type”, Type ‘mixed type”

* Sequence number

“design method” shall be Marshall

“aggregate type” shall be either stone and gravel

“mix type” from Table 401.4.2A